Knapsack trace: inputs and method

Investment Analyst: **Joan** · Emirates Post Network Audit · Tab 3 · Page 2 holds the completed grid, page 3 the traceback and the greedy comparison. The same input in Tab 3 (expander 'Knapsack grid in AED 10,000 units') and in Knapsack_Trace_Joan.ipynb gives the same grid and answer.

Question

Promise 20 min from the Al Ramool hub; four offices fail (Dijkstra, Tab 2). Budget **AED 450,000** for year one, set by Joan. Which upgrades give the largest yearly benefit without going over budget?

Inputs (rows in the group's matrix order: UMS, JAF, SHJ, AJM)

Row	Office	Upgrade	Cost (AED)	Cost in AED 10,000 units	Benefit (AED/yr)	Benefit (AED '000)	Benefit per AED
1	UMS	Express motorbike courier	69,900	7	135,000	135	1.93
2	JAF	JAFZA mini-hub	143,520	14	360,000	360	2.51
3	SHJ	Sharjah mini-hub	130,920	13	225,000	225	1.72
4	AJM	Ajman mini-hub	137,520	14	180,000	180	1.31
	Budget		450,000	45			

Rule for every cell

K[0][w] = 0 for every w. For row i and budget w: if $cost_i$ > w, then K[i][w] = K[i-1][w] (cannot afford it). Otherwise K[i][w] = max(K[i-1][w] (leave it), $benefit_i$ + K[i-1][w - $cost_i$] (take it)).

Traceback

Start at K[4][45]. If the cell differs from the one directly above, that upgrade was taken: move left by its cost. If it is the same, the upgrade was left: go up a row. Stop at row 0.

Greedy comparison

Sort by benefit per AED (highest first). Take each upgrade that still fits in the money left, then compare its total with the DP answer.

Rounding note: costs are rounded to the nearest AED 10,000 only to keep the grid to 46 columns. Tab 3 computes in exact AED and shows that both give the same choice at AED 450,000.

Completed knapsack grid (Joan)

Cells = best yearly benefit in AED '000 using the first i upgrades with budget w (units of AED 10,000). Highlighted cells = traceback path. Bottom-right K[4][45] = **765**.

row / budget w	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22
0 (none)	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1 UMS (c=7, b=135)	0	0	0	0	**0**	0	0	135	135	135	135	135	135	135	135	135	135	135	135	135	135	135	135
2 JAF (c=14, b=360)	0	0	0	0	0	0	0	135	135	135	135	135	135	135	360	360	360	360	**360**	360	360	495	495
3 SHJ (c=13, b=225)	0	0	0	0	0	0	0	135	135	135	135	135	135	225	360	360	360	360	360	360	360	495	495
4 AJM (c=14, b=180)	0	0	0	0	0	0	0	135	135	135	135	135	135	225	360	360	360	360	360	360	360	495	495

row / budget w	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
0 (none)	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1 UMS (c=7, b=135)	135	135	135	135	135	135	135	135	135	135	135	135	135	135	135	135	135	135	135	135	135	135	135
2 JAF (c=14, b=360)	495	495	495	495	495	495	495	495	495	495	495	495	495	495	495	495	495	495	495	495	495	495	495
3 SHJ (c=13, b=225)	495	495	495	495	585	585	585	585	**585**	585	585	720	720	720	720	720	720	720	720	720	720	720	720
4 AJM (c=14, b=180)	495	495	495	495	585	585	585	585	585	585	585	720	720	720	720	720	720	720	765	765	765	765	**765**

Traceback: K[4][45]=765 ≠ 720: take AJM (w 45→31) · K[3][31]=585 ≠ 495: take SHJ (w 31→18) · K[2][18]=360 ≠ 135: take JAF (w 18→4) · K[1][4]=0 = 0: leave UMS

Chosen: JAF + SHJ + AJM **Best benefit:** 765 (AED 765,000/yr) **Units used:** 41 of 45

Greedy: order JAF > UMS > SHJ > AJM picks JAF + UMS + SHJ total 720 **Gap (DP - greedy):** 45 (AED 45,000/yr) **Trace by:** Joan (Investment Analyst)

Traceback and greedy comparison

Start at the bottom-right cell; a cell that differs from the one above means that upgrade was taken.

Step	Cell	Compare with cell above	Decision	Budget left (units)
1	K[4][45] = 765	K[3][45] = 720	TAKE AJM	31
2	K[3][31] = 585	K[2][31] = 495	TAKE SHJ	18
3	K[2][18] = 360	K[1][18] = 135	TAKE JAF	4
4	K[1][4] = 0	K[0][4] = 0	leave UMS	4

Greedy order	Ratio	Cost (units)	Fits?	Left after
JAF	2.51	14	yes	31
UMS	1.93	7	yes	24
SHJ	1.72	13	yes	11
AJM	1.31	14	no	11

Result

	Dynamic programming	Greedy
Upgrades funded	JAF + SHJ + AJM mini-hubs	JAF + UMS + SHJ
Spend (exact AED)	411,960	344,340
Budget left (exact AED)	38,040	105,660
Yearly benefit (AED)	765,000	720,000
Offices meeting the 20-min promise	5 of 6	5 of 6

Greedy loses AED 45,000 a year. It takes JAF first (best ratio) and then the cheap UMS motorbike, leaving 11 units: too little for a 14-unit mini-hub. The DP skips UMS so two more mini-hubs fit.

Agreement with the dashboard: Tab 3 at budget AED 450,000 and promise 20 min shows the same grid, the same bottom-right value (765) and the same chosen upgrades. The exact AED calculation (no rounding) and a brute-force check of all 16 combinations give the same answer.